KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 21 · MATH

The Yellowstone Trail Map

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Finding Your Way

Quinlan is spending a week in Yellowstone National Park with his cousin Ramona. Warm up before the lesson and show your work. Some problems may feel new, and that is okay!

1. Review (Unit 4, volume): a) A storage bin is 7 feet long, 5 feet wide, and 2 feet tall. What is its volume? b) A box holds 240 cubic inches. Its base is 10 inches long and 6 inches wide. How tall is it?

2. Start at 0 and count by 4s. Write the first five numbers. Then start at 0 and count by 8s. Write the first five numbers.

3. Look at your two lists. Compare the numbers in the same position (the 2nd and the 2nd, the 3rd and the 3rd). What do you notice?

4. A trail map says the campsite is at (3, 5). What do you think the two numbers tell you to do?

   Move 3 right, then 5 up | Move 3 up, then 5 right | Move 8 steps in any direction

5. Draw two number lines from 0 to 6 that meet at 0, one going right and one going up. Mark a dot 2 steps right and 4 steps up.

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Ranger Roderick's Map

Walk right for the x-coordinate, then climb up for the y-coordinate. Show your work.

1. a) Write the ordered pair for a point 7 units right and 2 units up. b) Is (0, 6) on the x-axis or the y-axis? c) Start at (5, 4). Move 3 left and 4 up. Where do you stop?

2. Two steps: the Ranger Station is at (2, 3) and the Overlook is at (2, 9). Each grid unit stands for 1/4 mile. How many miles apart are they? How many miles is a trip there and back?

3. Rule A: start at 0, add 6. Rule B: start at 0, add 3. Write the first 5 numbers of each rule, then the ordered pairs (Rule A, Rule B). What is the relationship?

4. Ramona graphs (hours hiked, miles walked): (1, 3), (2, 6), (3, 9). If the pattern continues, which point shows 7 hours?

   (7, 21) (21, 7) (7, 12)

5. Find the mistake: Ruben says (4, 1) and (1, 4) are the same point because they use the same numbers. Explain his mistake in two or three complete sentences.

★ **Challenge:** Rule A: start at 1, add 2. Rule B: start at 4, add 2. Write four ordered pairs. Is each Rule B number "3 times" or "3 more" than Rule A? How do you know?

Looking Ahead: Week 22: Ship Day in Duluth Harbor. Convert customary and metric units and solve multistep problems.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 21 · READING

Homes in the Canyon Walls

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: How Is a Text Built?

Today you will learn how writers organize facts, using articles about Mesa Verde in Colorado. Show what you already know. It is fine to guess!

1. Review (Week 19): Read this Aesop fable: A lion let a tiny mouse go free. Later, the mouse chewed through a hunter's net and freed the lion. Write the **theme** of the fable as one full sentence.

2. Read: *First, we packed the car. Next, we drove to the canyon. Finally, we hiked to the overlook.* Which three words tell you the order of events?

3. Which word usually tells WHY something happened?

   because first both

4. Long ago, families at Mesa Verde built stone homes inside shallow caves in the cliff walls. Write one reason someone might build a home in a cliff wall.

5. You want to explain your favorite sport to a friend. Would you tell it in time order, compare it with another sport, or describe its parts? Explain your choice in one or two sentences.

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Homes in the Canyon Walls

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Structure Spotter

Write in complete sentences with correct capital letters, commas, and verbs.

1. Read: *Farmers had very little water. To fix this, they stored rain in reservoirs and built homes near springs.* Name the text structure, and copy one signal phrase that proves it.

2. Be the editor. Rewrite each sentence correctly. a) In the late 1200s a long drought struck. b) the zuni and hopi people live in new mexico and arizona. c) The kivas at Cliff Palace was used for ceremonies.

3. Find the mistake! Zane says, “Text A is cause and effect because it uses the word *Then*.” Which explanation is correct?

   - He is wrong: Text A is chronology, and one signal word is not proof.
   - He is right: any text with “Then” is cause and effect.
   - He is wrong: Text A has no structure.

4. Writing: In 4 or 5 sentences, compare a pithouse and a cliff dwelling. Use the signal words **both** and **however**, and put a comma after every introductory word or phrase.

★ **Challenge:** Write the drought facts two ways: as chronology (time words), then as cause and effect (because, as a result).

Looking Ahead: Week 22: Firsthand and Secondhand Accounts. You will compare the Apollo 11 astronauts’ own words from the Moon with an encyclopedia-style article about the 1969 landing.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 21 · SCIENCE

Physical and Chemical Changes: Where Does the Matter Go?

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 19):** Undisturbed rock layers, top to bottom, are sandstone with tracks, limestone with corals and shale with fern prints. Which layer formed first? What environment did the limestone form in?

2. What do you think is the difference between a physical change and a chemical change? Give one example of each.

3. Predict: A candle burns on a scale for an hour. Will the scale show more, less or the same mass? Where do you think the missing wax goes?

4. Circle the changes you think make a NEW substance: ice melting · toast turning brown · a nail rusting · sugar dissolving · paper burning

5. Draw what you think happens to the tiny particles of a substance during a chemical change. Label your drawing.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 21 · SCIENCE

Physical and Chemical Changes: Where Does the Matter Go?

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Account for Every Gram

Use evidence from today's lesson. Write in complete sentences.

1. Bianca sees bubbles in two cups: water boiling on the stove, and cold vinegar with baking soda. Which cup shows a chemical change? Explain using the words **new substance**.

2. **Two-step problem.** An open dish of reactants weighs 86.4 grams. After the reaction it weighs 82.9 grams. How many grams of gas escaped? What would a sealed jar show?

3. **Find the mistake.** Boaz says: "Burning steel wool gets heavier, so it creates new matter." Explain.

4. Why did the Statue of Liberty's copper stay strong while an iron gate crumbled? Use the word **patina**.

★ **Challenge:** Draw 8 toy bricks as two reactants, then the same bricks rebuilt as one product. Explain how this models conservation of mass.

Looking Ahead: Next week: Senses and Signals. How do animals sense the world, send signals through nerves to the brain and respond?

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 21 · SOCIAL STUDIES

World War I and the Great Migration (1914–1920s)

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Write what you know right now. Guesses are fine; this is your starting point.

1. **Review (Week 19):** The 19th Amendment (1920) protected women's votes. Name one group of women who were still kept from voting after 1920, and explain why.

2. What do you already know about World War I? Write anything: when, where, or who fought.

3. Prediction: Why might a country that stayed out of a war for years finally decide to join it?

4. Why do you think a family might pack up and move to a different part of the country? Give two reasons.

5. Write one question you have about soldiers, workers or families during World War I.

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World War I and the Great Migration (1914–1920s)

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Democracy Abroad, Democracy at Home

Use evidence from today's lesson. Write in complete sentences.

1. Put in order with years: the armistice; the Lusitania sinks; the United States declares war; the 369th marches up Fifth Avenue.

2. **Two steps:** Detroit had 5,741 Black residents in 1910 and 40,838 in 1920. How many more? Show your work and give one reason.

3. **Find the mistake.** "Every soldier who served in World War I was treated the same and honored right away." Rewrite it so it is true, using one person from the lesson.

4. **Compare two sources.** Wilson said in 1917, "The world must be made safe for democracy." Du Bois wrote in 1919, "We return fighting." How does Du Bois answer Wilson?

★ **Challenge:** Write a 2-sentence letter from a family member in Detroit in 1920 to a cousin in the South: what is better, and what is still hard?

Looking Ahead: Next week: The 1920s and the Harlem Renaissance, with Langston Hughes, Zora Neale Hurston, Duke Ellington, Louis Armstrong and the true story of Greenwood in Tulsa.